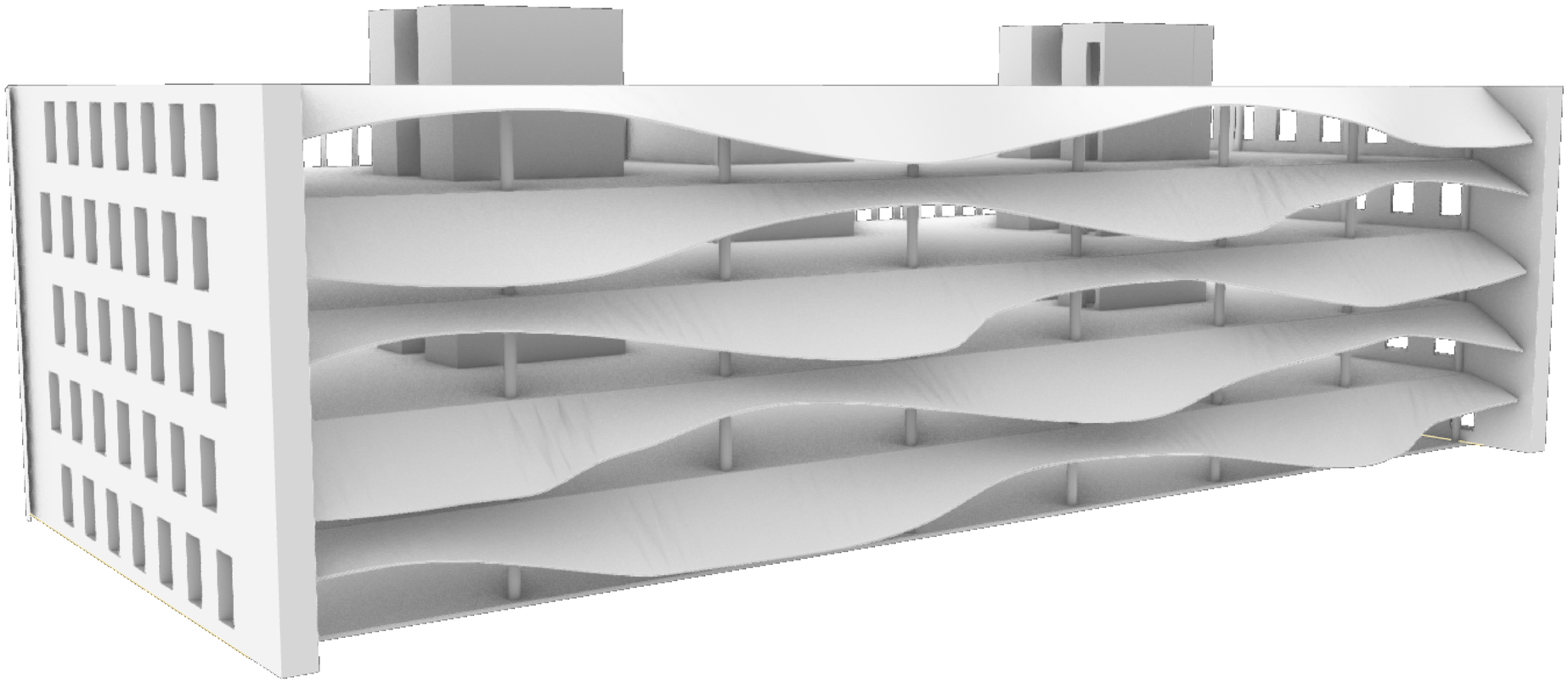


Sheet List	
Sheet Number	Sheet Name
A001	TITLE SHEET
A002	SITE INFORMATION
A100	TYPICAL FLOOR PLAN
A101	REFLECTED CEILING PLAN
A200	NORTH AND EAST ELEVATIONS
A201	SOUTH AND WEST ELEVATIONS
A300	BUILDING SECTION
A400	WALL SECTION
A500	ENLARGED TOILET PLAN & ELEVATIONS
A600	PERFORMANCE ANALYSIS
A601	PERFORMANCE ANALYSIS
E100	ELECTRICAL POWER PLAN
FS100	FIRE SAFETY PLAN
M100	TYPICAL MECHANICAL PLAN

Phoenix Office Building

100 S Rural Rd Tempe, AZ 85288

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No.	Description	Date

90% CD SET

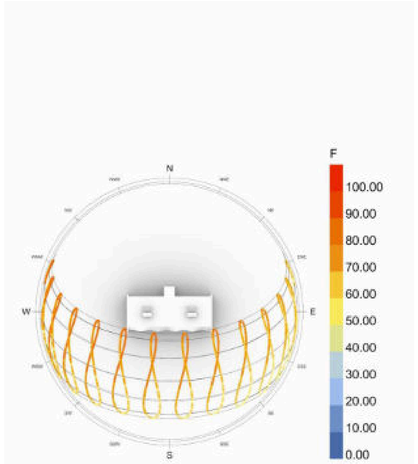
TITLE SHEET

Date	03.05.24
Drawn By	Author
Checked By	Checker

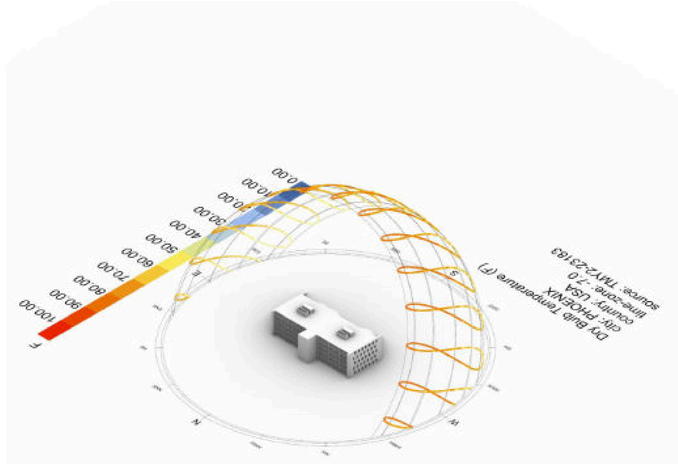
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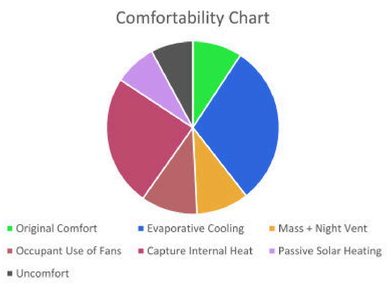
Scale

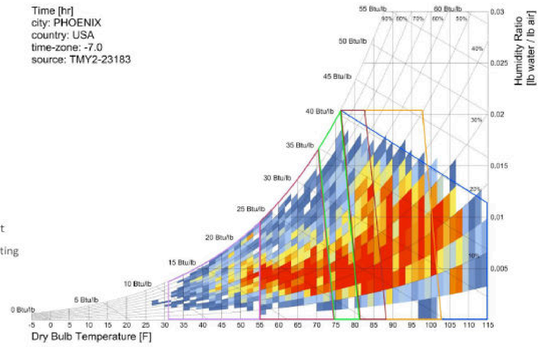


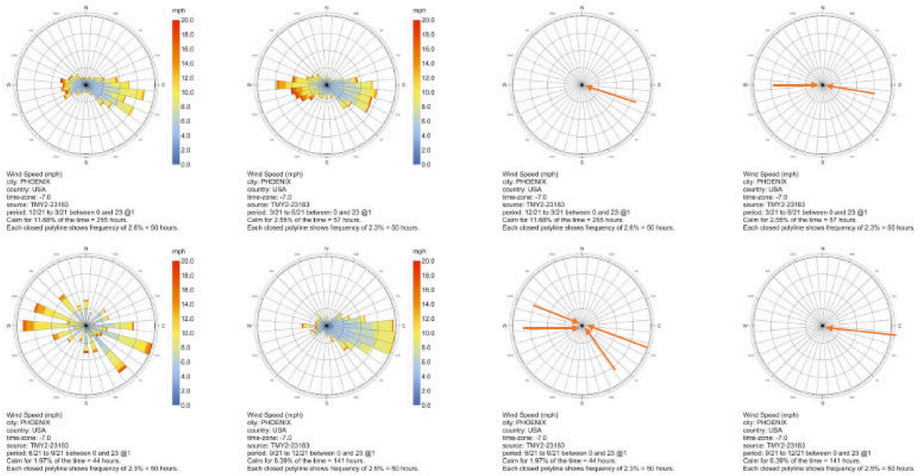


Dry Bulb Temperature (F)
city: PHOENIX
country: USA
time-zone: -7.0
source: TMY2-23183









Phoenix Office Building

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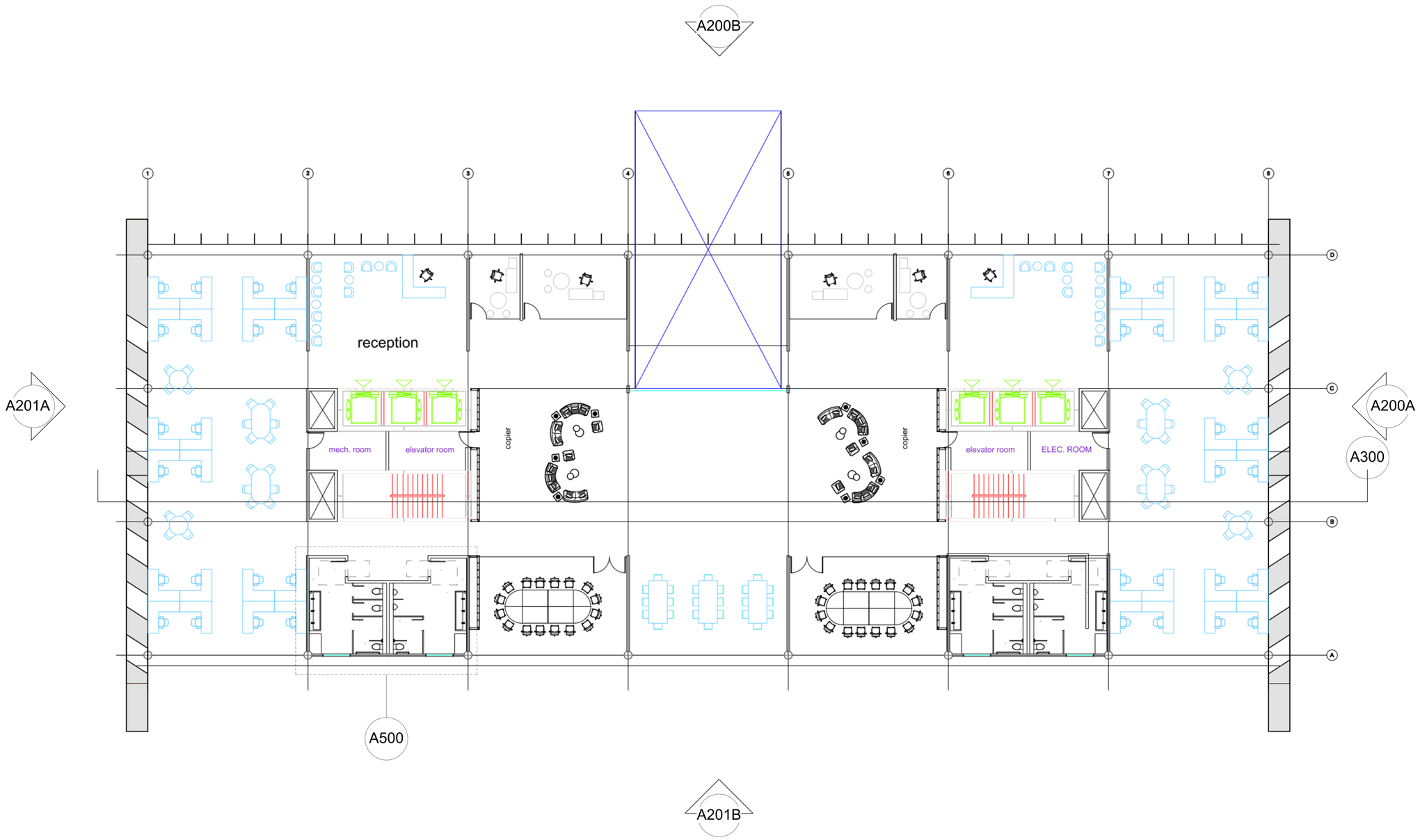
90% CD SET

SITE INFORMATION

Date	03.05.24
Drawn By	Author
Checked By	Checker

A002

Scale



Phoenix Office Building

100 S Rural Rd Tempe, AZ 85288

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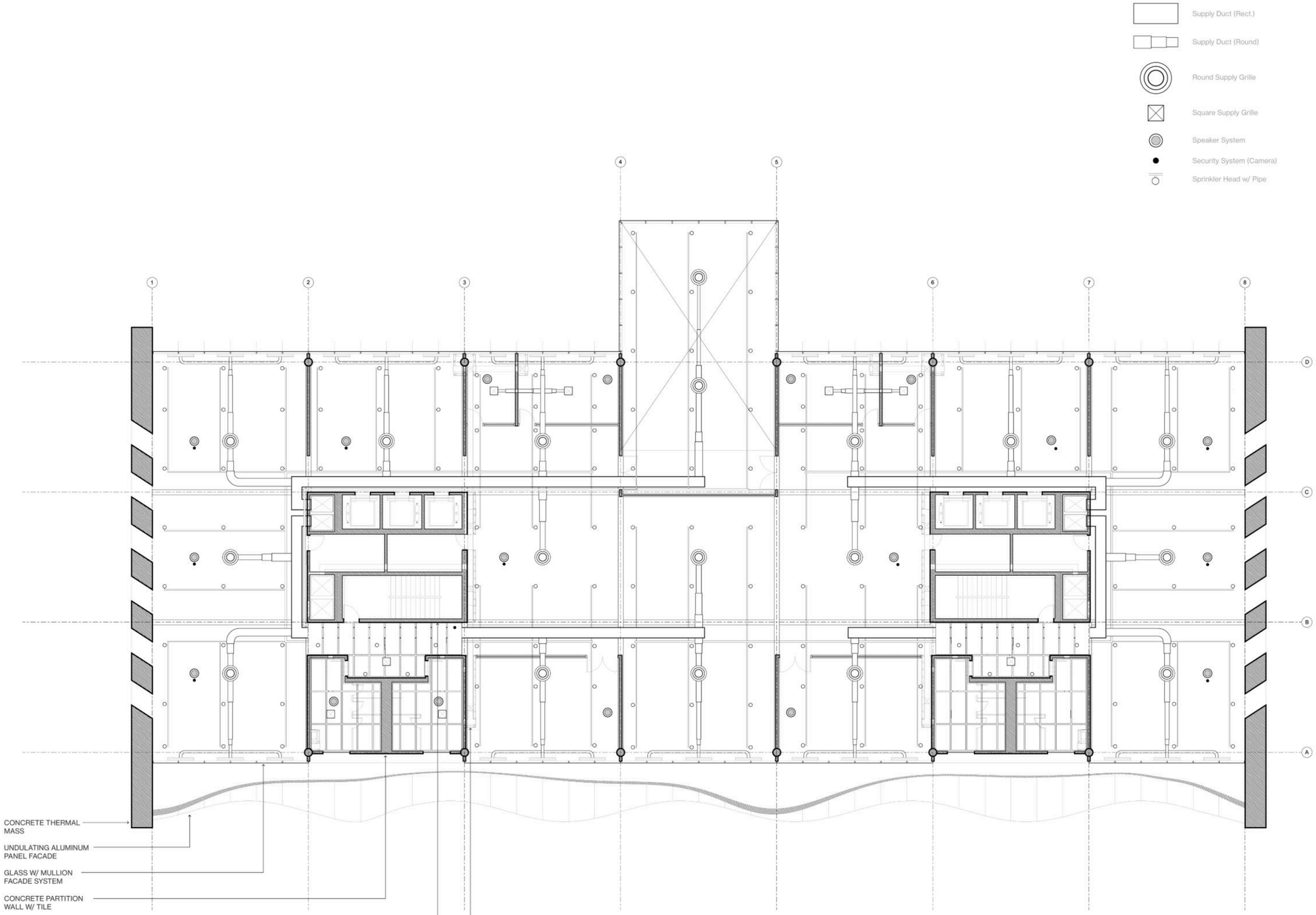
90% CD SET

TYPICAL FLOOR PLAN

Date	03.05.24
Drawn By	Author
Checked By	Checker

A100

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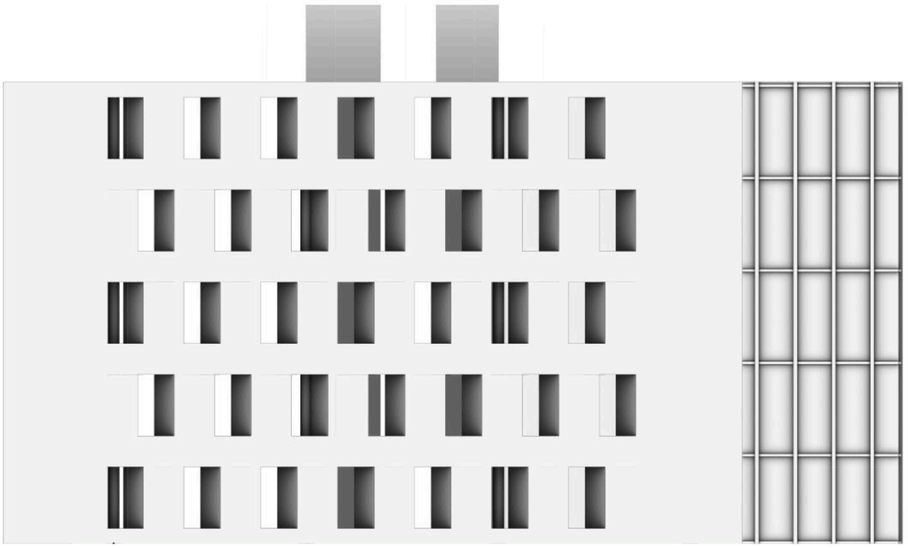
90% CD SET

REFLECTED CEILING PLAN

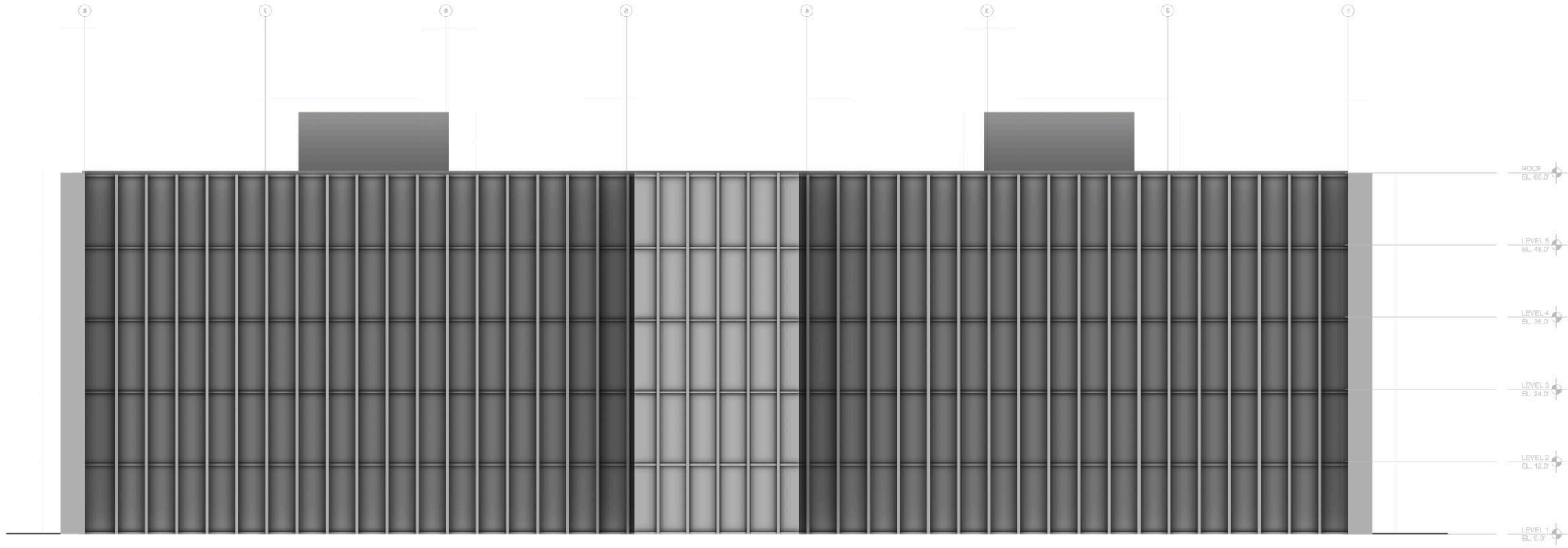
Date	03.05.24
Drawn By	Author
Checked By	Checker

A101

Scale



A200A



A200B

Phoenix Office Building

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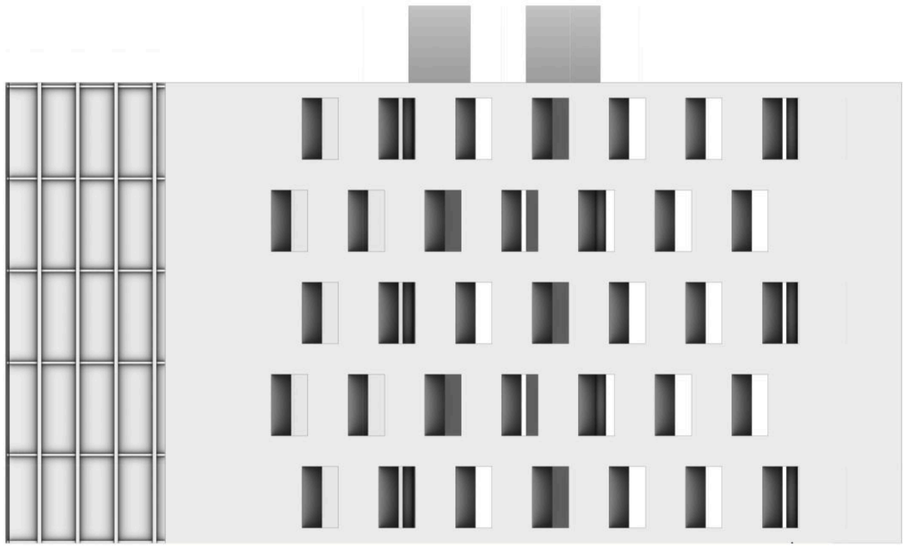
90% CD SET

NORTH AND EAST ELEVATIONS

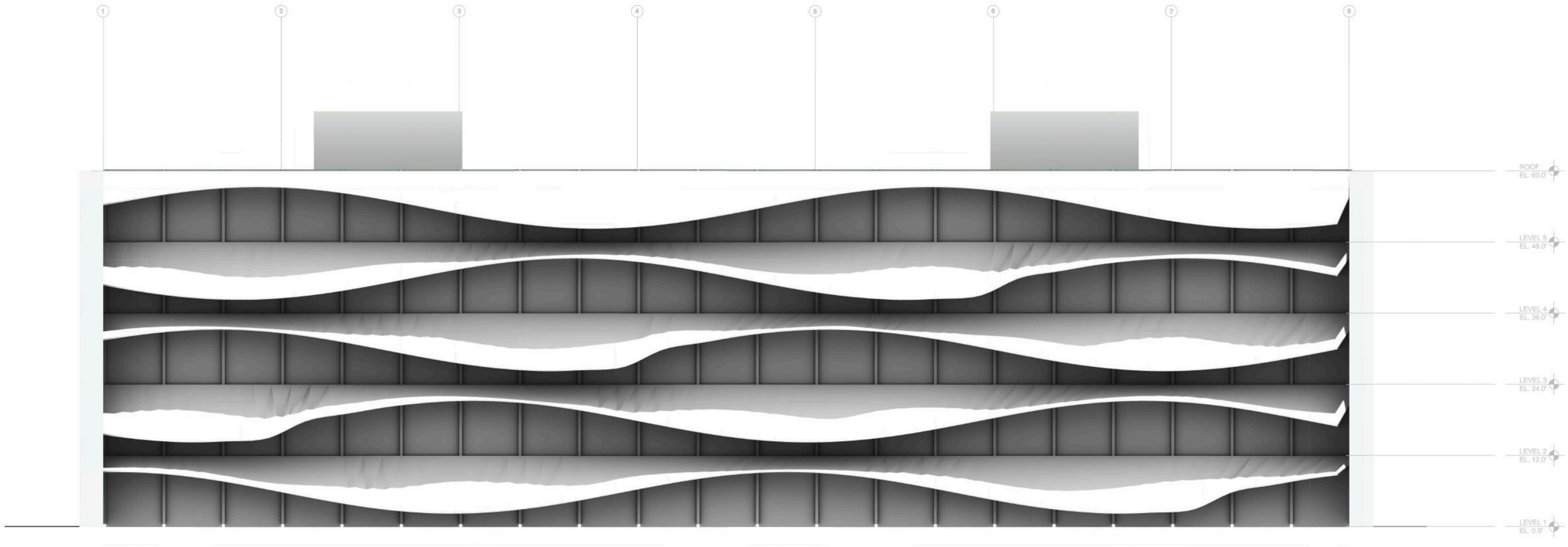
Date	03.05.24
Drawn By	Author
Checked By	Checker

A200

Scale



A201A



A201B

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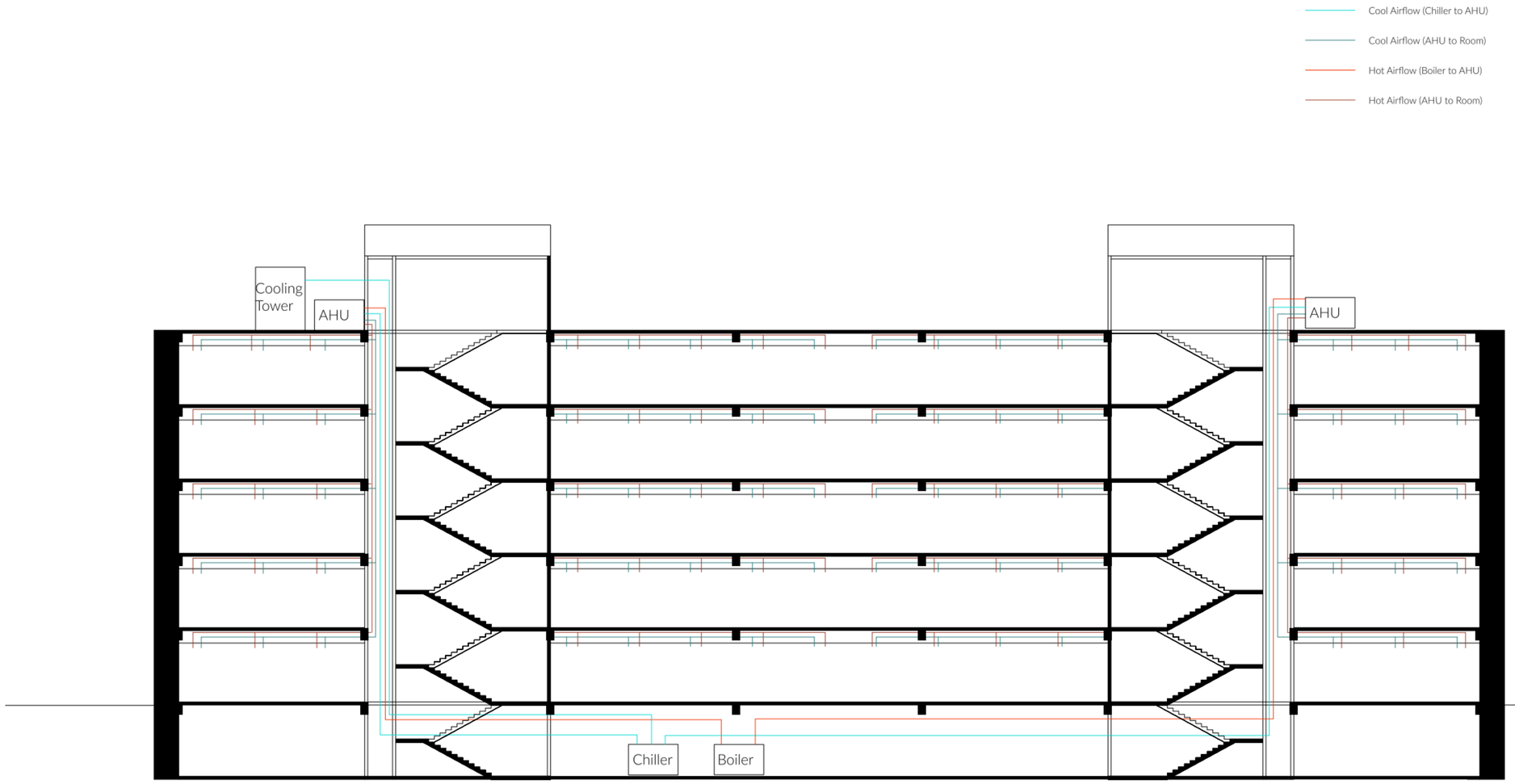
90% CD SET

SOUTH AND WEST ELEVATIONS

Date	03.05.24
Drawn By	Author
Checked By	Checker

A201

Scale



Ventilation Section @ 1/2" Scale

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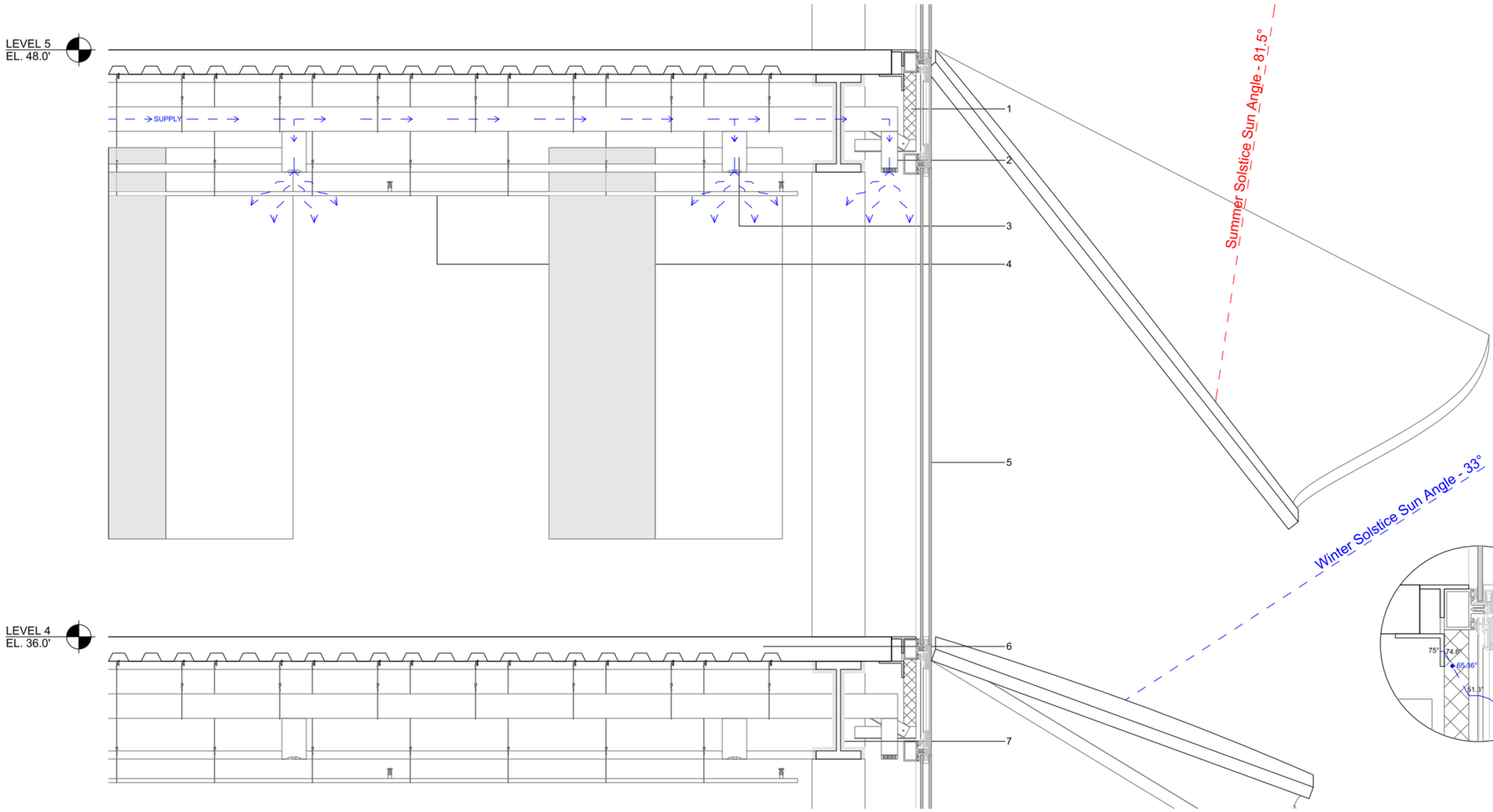
90% CD SET

BUILDING SECTION

Date	03.05.24
Drawn By	Author
Checked By	Checker

A300

Scale



1 — XPS insulation
2 — Slot Diffusers
3 — Round Grills
4 — Sprinklers
5 — Double Low-E Glass
6 — Concrete on Metal Deck
7 — Steel Beam

Dew Point Calculations:
Tdp = T - ((100-RH)/5)
Tdp – Temperature Dew Point
T – Temperature
RH – Relative Humidity

Summer Dew Point
T = 75
RH = 19.4%
Tdp = 75 - ((100-19.4)/5) = 58.88°F

Winter Dew Point
T = 75
RH = 51.8%
Tdp = 75 - ((100-51.8)/5) = 65.36°F

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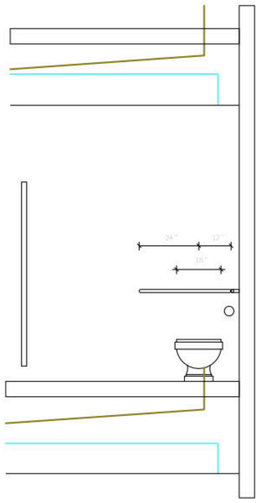
90% CD SET

WALL SECTION

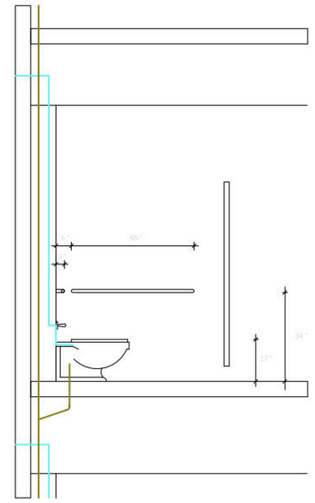
Date	03.05.24
Drawn By	Author
Checked By	Checker

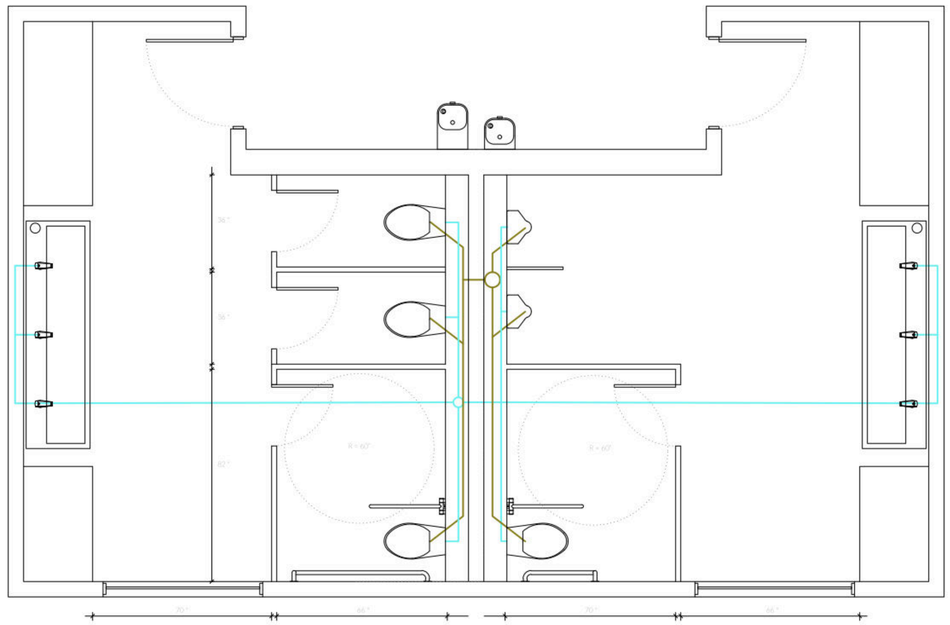
A400

Scale



ADA Compliant Restroom Section @ 1/2" Scale





Restroom Plan @ 1/2" Scale

Occupant Load Calculations

-Occupancy = Business
-Maximum Floor Area Allowances (Business Occupancy) = 100 Gross Sq. Ft.
-Total Floor Area (Sq. Ft.) = 12,799

Occupant Load per Floor = 12,799 Sq. Ft. / 100 Gross Sq. Ft.
= 127.99 = 127 Maximum Occupancy Load (Business)
+4 (1 Occupant per Service Room)
= 131 Total Occupant Load per Floor

131 Total Occupant Load / 2 = 65.5 = 66 Male Occ. + 66 Female Occ.

of Water Closets (1 per 25 for first 50, then 1 per 50 for remaining)
66 = 25 + 25 + 16 = 3 Minimum Water Closets (M/F) per Floor < 6 Closets (PASS)

of Lavatories (1 per 40 for first 80, then 1 per 50 for remaining)
66 = 40 + 26 = 2 Minimum Lavatories (M/F) per floor < 6 Lavatories (PASS)

of Drinking Fountains (1 per 100)
66 = 1 Minimum Drinking Fountain < 4 Drinking Fountains (PASS)

of Service Sinks = 1 Per Floor = 1 Service Sink Existing (PASS)

MAXIMUM FLOOR AREA ALLOWANCES PER OCCUPANT

OCCUPANCY	FLOOR AREA IN SQ. FT PER OCCUPANT
Agricultural building	300 gross
Aircraft hangars	500 gross
Airport terminal Baggage claim Baggage handling Concourse Waiting areas	20 gross 300 gross 100 gross 15 gross
Assembly Gaming floors (keno, slots, etc.)	11 gross
Assembly with fixed seats	See Section 1005.7.2 ¶
Assembly without fixed seats Concentrated (chairs only—not fixed) Standing space Unconcentrated (tables and chairs)	7 net 5 net 15 net
Bowling centers, allow 5 persons for each lane including 15 feet of runway, and for additional	7 net
Business areas	100 gross
Conference—other than fixed seating areas	10 net
Dormitories	50 gross
Educational Classroom area Shops and other vocational room areas	20 net 50 net
Exercise rooms	50 gross
H-5 Fabrication and manufacturing areas	200 gross
Industrial areas	100 gross
Institutional areas Inpatient treatment areas Outpatient areas Sleeping areas	240 gross 100 gross 120 gross
Kitchens, commercial	200 gross
Library Reading rooms Stack area	50 net 100 gross
Locker rooms	50 gross
Mercantile Areas on other floors Basement and grade floor areas Storage, stock, shipping areas	60 gross 30 gross 300 gross
Parking garages	200 gross
Residential	200 gross
Skating rinks, swimming pools Rink and pool Decks	50 gross 15 gross
Stages and platforms	15 net
Accessory storage areas, mechanical equipment room	300 gross
Warehouses	500 gross

For SI: 1 square foot = 0.0929 m².

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Phoenix Office Building

100 S Rural Rd Tempe, AZ 85288

No.	Description	Date

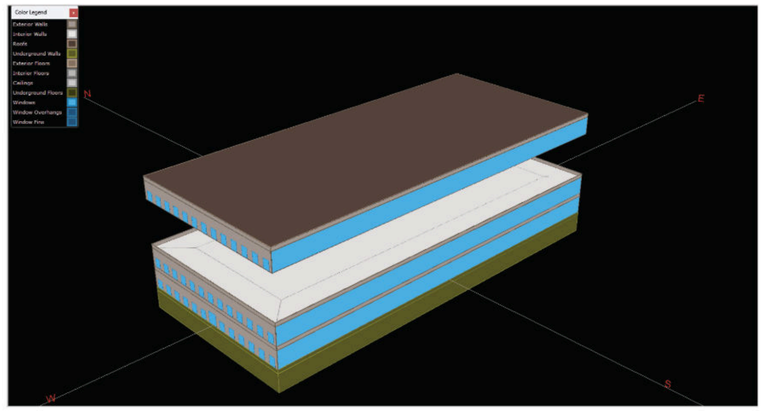
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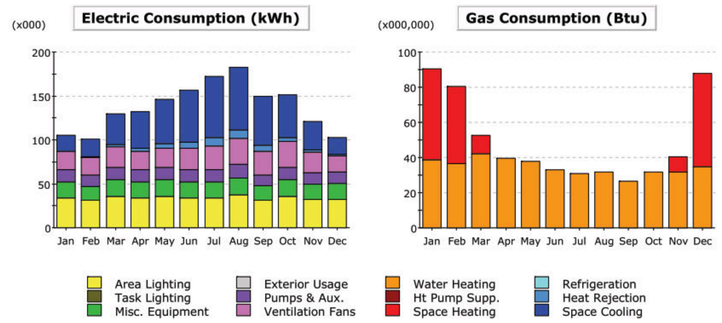
ENLARGED TOILET PLAN & ELEVATIONS

Date	03.05.24
Drawn By	Author
Checked By	Checker

A500

Scale





Electric Consumption (kWh x000)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Space Cool	17.8	19.9	35.3	41.4	50.6	58.9	69.2	70.9	55.4	48.6	32.7	19.6	520.2
Heat Reject.	0.7	0.9	2.6	3.5	4.9	6.9	9.6	9.6	7.0	4.7	2.5	1.0	54.0
Refrigeration	-	-	-	-	-	-	-	-	-	-	-	-	-
Space Heat	-	-	-	-	-	-	-	-	-	-	-	-	-
HP Supp.	-	-	-	-	-	-	-	-	-	-	-	-	-
Hot Water	-	-	-	-	-	-	-	-	-	-	-	-	-
Vent. Fans	20.5	19.9	22.9	21.3	21.7	24.6	27.0	29.8	26.9	28.8	22.8	18.9	285.1
Pumps & Aux.	13.8	12.5	14.5	13.8	14.5	13.8	13.8	15.2	12.5	14.5	13.2	13.2	165.5
Ext. Usage	-	-	-	-	-	-	-	-	-	-	-	-	-
Misc. Equip.	18.2	16.4	18.9	18.1	18.9	18.0	18.2	19.6	16.6	18.9	17.3	17.4	216.4
Task Lights	-	-	-	-	-	-	-	-	-	-	-	-	-
Area Lights	34.1	30.9	35.7	34.1	35.7	34.1	34.1	37.2	31.0	35.7	32.5	32.6	407.6
Total	105.1	100.6	129.8	132.2	146.3	156.4	171.9	182.2	149.4	151.2	121.1	102.7	1,648.8

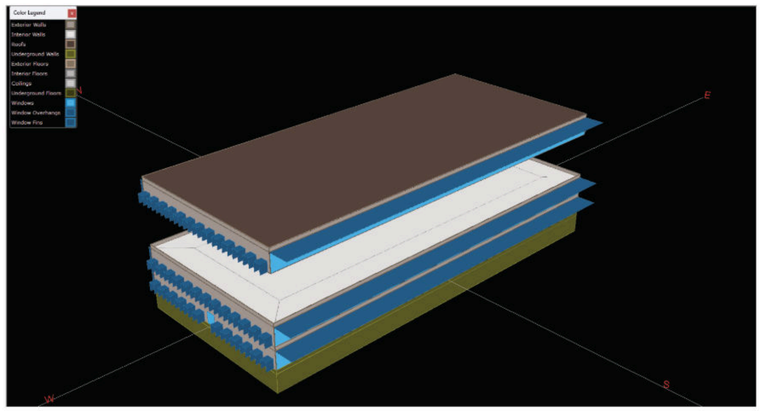
Gas Consumption (Btu x000,000)

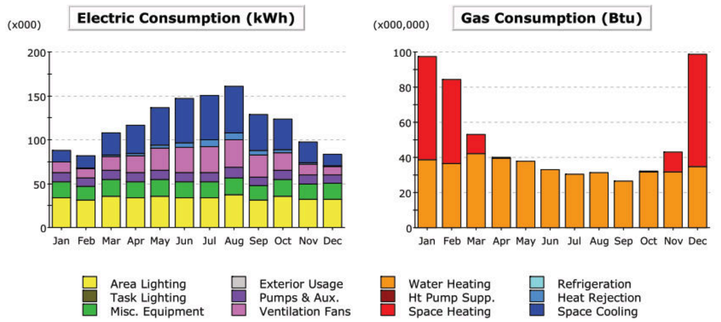
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Space Cool	-	-	-	-	-	-	-	-	-	-	-	-	-
Heat Reject.	-	-	-	-	-	-	-	-	-	-	-	-	-
Refrigeration	-	-	-	-	-	-	-	-	-	-	-	-	-
Space Heat	51.74	43.95	10.60	0.35	-	-	-	-	-	-	9.00	52.98	168.62
HP Supp.	-	-	-	-	-	-	-	-	-	-	-	-	-
Hot Water	38.76	36.46	42.01	39.37	37.99	33.23	30.70	31.55	26.49	31.71	31.58	34.66	414.50
Vent. Fans	-	-	-	-	-	-	-	-	-	-	-	-	-
Pumps & Aux.	-	-	-	-	-	-	-	-	-	-	-	-	-
Ext. Usage	-	-	-	-	-	-	-	-	-	-	-	-	-
Misc. Equip.	-	-	-	-	-	-	-	-	-	-	-	-	-
Task Lights	-	-	-	-	-	-	-	-	-	-	-	-	-
Area Lights	-	-	-	-	-	-	-	-	-	-	-	-	-
Total	90.50	80.41	52.61	39.71	37.99	33.23	30.70	31.55	26.49	31.71	40.58	87.64	583.13

System & Zone Name	System Type Principal Zone Activity	Type* Ret Zn	Area sqft	Design Flow Supply cfm	Supply cfm/sf	Min Flow	Design Ventilation OSA cfm	OSA %	OSA cfm/sf	OSA cfm/per	Design Capacity Cool tons	Cool sf/ton	Cool cfm/ton	Cool Btuh/sf	Heat Btuh/sf	Hrs Outside Thrt-Range Cool Hrs	Heat Hrs
Sys1 (VAVS) (T)	Variable Air Volume	D	20,928	34,909	1.67	32%	4,763	14%	0.228	22.8	95	220	367	54.5	11.3	4,519	0
··· South Perim Zn (T.S19)	Office (Open Plan)	C	3,045	15,912	5.23	30%	633	4%	0.208	20.8	43	70	367	170.7	35.3	390	0
··· East Perim Zn (T.E20)	Office (Open Plan)	C	1,215	2,421	1.99	30%	253	10%	0.208	20.8	7	184	367	65.1	13.4	957	0
··· North Perim Zn (T.N21)	Office (Open Plan)	C	3,045	6,685	2.20	30%	633	9%	0.208	20.8	18	167	367	71.7	14.8	653	0
··· West Perim Zn (T.W22)	Office (Open Plan)	C	1,215	2,472	2.03	30%	253	10%	0.208	20.8	7	181	367	66.5	13.7	530	0
··· Core Zn (T.C23)	Office (Open Plan) (51%)	C	12,408	7,420	0.60	40%	2,991	40%	0.241	24.1	20	614	367	19.5	4.0	1,989	0
··· Pl Zn (T.24)	(unknown)	U	20,928	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Sum of Zones		··	··	34,909	··	··	··	··	··	··	95	··	··	··	··	164%	0%
Sum of Zones / System Total		··	··	100%	··	··	··	··	··	··	100%	··	··	··	··	··	··

Project Totals System & Zone Name	System Type Principal Zone Activity	Type* Ret Zn	Area sqft	Design Flow Supply cfm	Supply cfm/sf	Min Flow	Design Ventilation OSA cfm	OSA %	OSA cfm/sf	OSA cfm/per	Design Capacity Cool tons	Cool sf/ton	Cool cfm/ton	Cool Btuh/sf	Heat Btuh/sf	Throttling Range Hrs Outside Thrt-Range
Sum of SYSTEMs	··	··	125,568	207,266	1.65	33%	45,818	31%	0.365	20.7	570	220	363	54.5	11.1	92%
Sum of ZONEs	··	··	··	207,266	1.65	··	··	··	··	··	570	··	··	··	··	··
Sum of Zones / System Total	··	··	··	100%	··	··	··	··	··	··	100%	··	··	··	··	··

* Return Types: ·· 'P' = Plenum Return ·· 'D' = Ducted Return ·· 'd' = Direct return ·· (Plenum Zones are not shown on this report)
* Zone Types: ·· 'C' = Conditioned Zone ·· 'U' = Unconditioned Zone ·· 'S' = Slave Zone ·· (conditioned but no t-stat)





Electric Consumption (kWh x000)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Space Cool	12.7	13.9	25.6	32.3	42.5	50.5	51.2	53.5	41.1	35.0	22.9	13.6	394.7
Heat Reject.	0.4	0.6	1.8	2.8	4.2	5.7	7.2	7.2	5.3	3.4	1.7	0.6	40.8
Refrigeration	-	-	-	-	-	-	-	-	-	-	-	-	-
Space Heat	-	-	-	-	-	-	-	-	-	-	-	-	-
HP Supp.	-	-	-	-	-	-	-	-	-	-	-	-	-
Hot Water	-	-	-	-	-	-	-	-	-	-	-	-	-
Vent. Fans	11.7	10.4	15.2	19.1	24.6	28.4	29.6	32.2	25.5	19.5	12.6	9.6	238.4
Pumps & Aux.	10.5	9.5	11.0	10.5	11.0	10.5	10.5	11.5	9.5	11.0	10.0	10.0	125.2
Ext. Usage	-	-	-	-	-	-	-	-	-	-	-	-	-
Misc. Equip.	18.2	16.4	18.9	18.1	18.9	18.0	18.2	19.6	16.6	18.9	17.3	17.4	216.4
Task Lights	-	-	-	-	-	-	-	-	-	-	-	-	-
Area Lights	34.1	30.9	35.7	34.1	35.7	34.1	34.1	37.2	31.0	35.7	32.5	32.6	407.6
Total	87.6	81.7	108.1	116.8	136.7	147.2	150.8	161.2	128.9	123.4	97.0	83.7	1,423.0

Gas Consumption (Btu x000,000)

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Total
Space Cool	-	-	-	-	-	-	-	-	-	-	-	-	-
Heat Reject.	-	-	-	-	-	-	-	-	-	-	-	-	-
Refrigeration	-	-	-	-	-	-	-	-	-	-	-	-	-
Space Heat	58.59	47.76	10.71	0.76	-	-	-	-	-	0.18	11.11	63.83	192.92
HP Supp.	-	-	-	-	-	-	-	-	-	-	-	-	-
Hot Water	38.89	36.58	42.12	39.44	38.02	33.22	30.56	31.44	26.48	31.85	31.74	34.78	415.12
Vent. Fans	-	-	-	-	-	-	-	-	-	-	-	-	-
Pumps & Aux.	-	-	-	-	-	-	-	-	-	-	-	-	-
Ext. Usage	-	-	-	-	-	-	-	-	-	-	-	-	-
Misc. Equip.	-	-	-	-	-	-	-	-	-	-	-	-	-
Task Lights	-	-	-	-	-	-	-	-	-	-	-	-	-
Area Lights	-	-	-	-	-	-	-	-	-	-	-	-	-
Total	97.48	84.34	52.83	40.19	38.02	33.22	30.56	31.44	26.48	32.03	42.85	98.61	608.04

INCREASES DUE TO LESS SUNLIGHT HEATING SPACE. THIS CAN BE IMPROVED.

System & Zone Name	System Type Principal Zone Activity	Type* Ret Zn	Area sqft	Design Flow Supply cfm	Supply cfm/sf	Min Flow	Design Ventilation OSA cfm	OSA %	OSA cfm/sf	OSA cfm/per	Design Capacity Cool tons	Cool sf/ton	Cool cfm/ton	Cool Btuh/sf	Heat Btuh/sf	Hrs Outside Thrt-Range Cool Hrs	Heat Hrs
Sys1 (VAVS) (T)	Variable Air Volume	D	20,928	24,282	1.16	33%	4,763	20%	0.228	22.8	69	305	354	39.3	7.8	5,480	2
··· South Perim Zn (T.S19)	Office (Open Plan)	C	3,045	7,118	2.34	30%	633	9%	0.208	20.8	20	151	354	79.2	15.8	785	0
··· East Perim Zn (T.E20)	Office (Open Plan)	C	1,215	1,485	1.22	30%	253	17%	0.208	20.8	4	290	354	41.4	8.3	1,232	1
··· North Perim Zn (T.N21)	Office (Open Plan)	C	3,045	6,684	2.20	30%	633	9%	0.208	20.8	19	161	354	74.4	14.8	880	0
··· West Perim Zn (T.W22)	Office (Open Plan)	C	1,215	1,575	1.30	30%	253	16%	0.208	20.8	4	273	354	44.0	8.8	1,048	1
··· Core Zn (T.C23)	Office (Open Plan) (51%)	C	12,408	7,420	0.60	40%	2,991	40%	0.241	24.1	21	592	354	20.3	4.0	1,535	0
··· Pl Zn (T.24)	(unknown)	U	20,928	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a	n/a
Sum of Zones		··	··	24,282	··	··	··	··	··	··	69	··	··	··	··	198%	0%
Sum of Zones / System Total		··	··	100%	··	··	··	··	··	··	100%	··	··	··	··	··	··

Project Totals System & Zone Name	System Type Principal Zone Activity	Type* Ret Zn	Area sqft	Design Flow Supply cfm	Supply cfm/sf	Min Flow	Design Ventilation OSA cfm	OSA %	OSA cfm/sf	OSA cfm/per	Design Capacity Cool tons	Cool sf/ton	Cool cfm/ton	Cool Btuh/sf	Heat Btuh/sf	Throttling Range Hrs Outside Thrt-Range
Sum of SYSTEMs	··	··	125,568	154,161	1.23	34%	45,818	41%	0.365	20.7	431	291	358	41.2	8.3	67%
Sum of ZONEs	··	··	··	154,161	1.23	··	··	··	··	··	431	··	··	··	··	··
Sum of Zones / System Total	··	··	··	100%	··	··	··	··	··	··	100%	··	··	··	··	··

* Return Types: ·· 'P' = Plenum Return ·· 'D' = Ducted Return ·· 'd' = Direct return ·· (Plenum Zones are not shown on this report)
* Zone Types: ·· 'C' = Conditioned Zone ·· 'U' = Unconditioned Zone ·· 'S' = Slave Zone ·· (conditioned but no t-stat)

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No.	Description	Date

90% CD SET

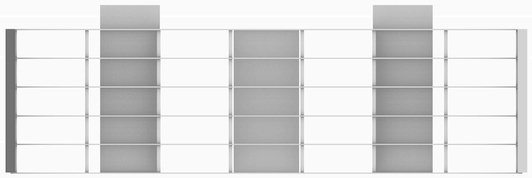
PERFORMANCE ANALYSIS

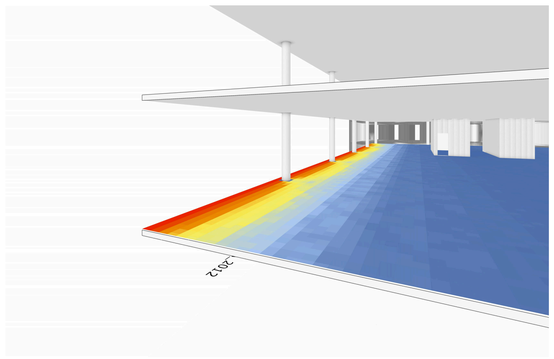
Date	03.05.24
Drawn By	Author
Checked By	Checker

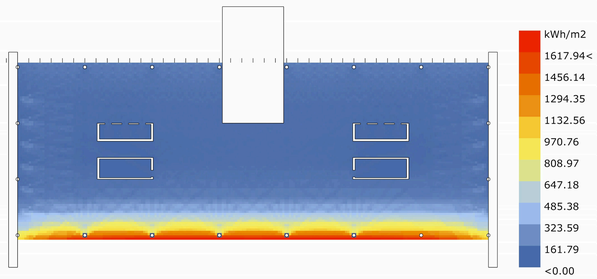
A600

Scale

ITERATION #1

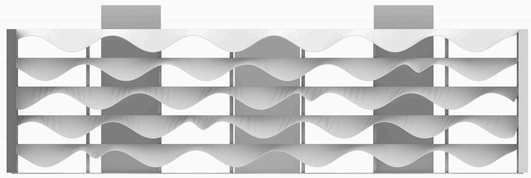


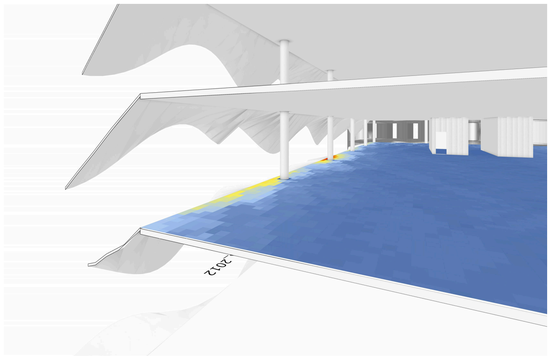


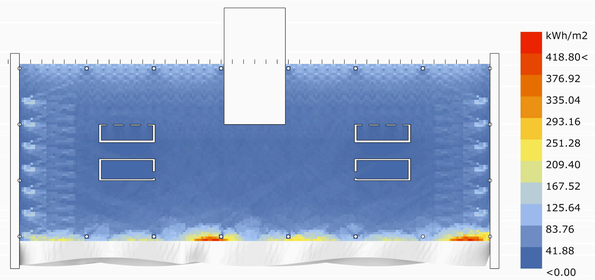


Radiation Analysis
Phoenix_Sky_Harbor_Intl_AP_AZ_USA_2012
1 JAN 1:00 - 31 DEC 24:00

ITERATION #2

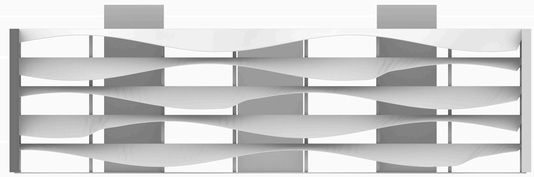


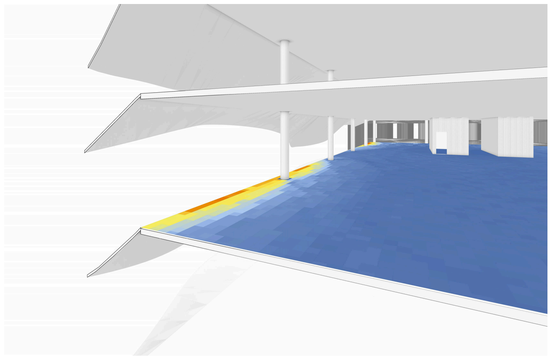


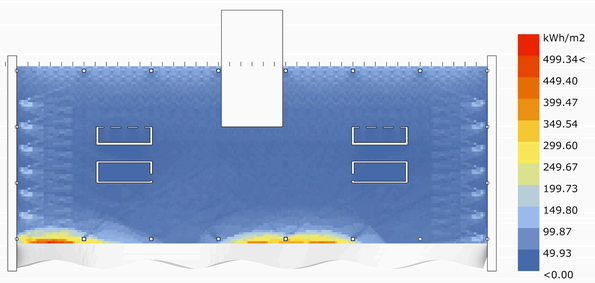


Radiation Analysis
Phoenix_Sky_Harbor_Intl_AP_AZ_USA_2012
1 JAN 1:00 - 31 DEC 24:00

ITERATION #3







Radiation Analysis
Phoenix_Sky_Harbor_Intl_AP_AZ_USA_2012
1 JAN 1:00 - 31 DEC 24:00

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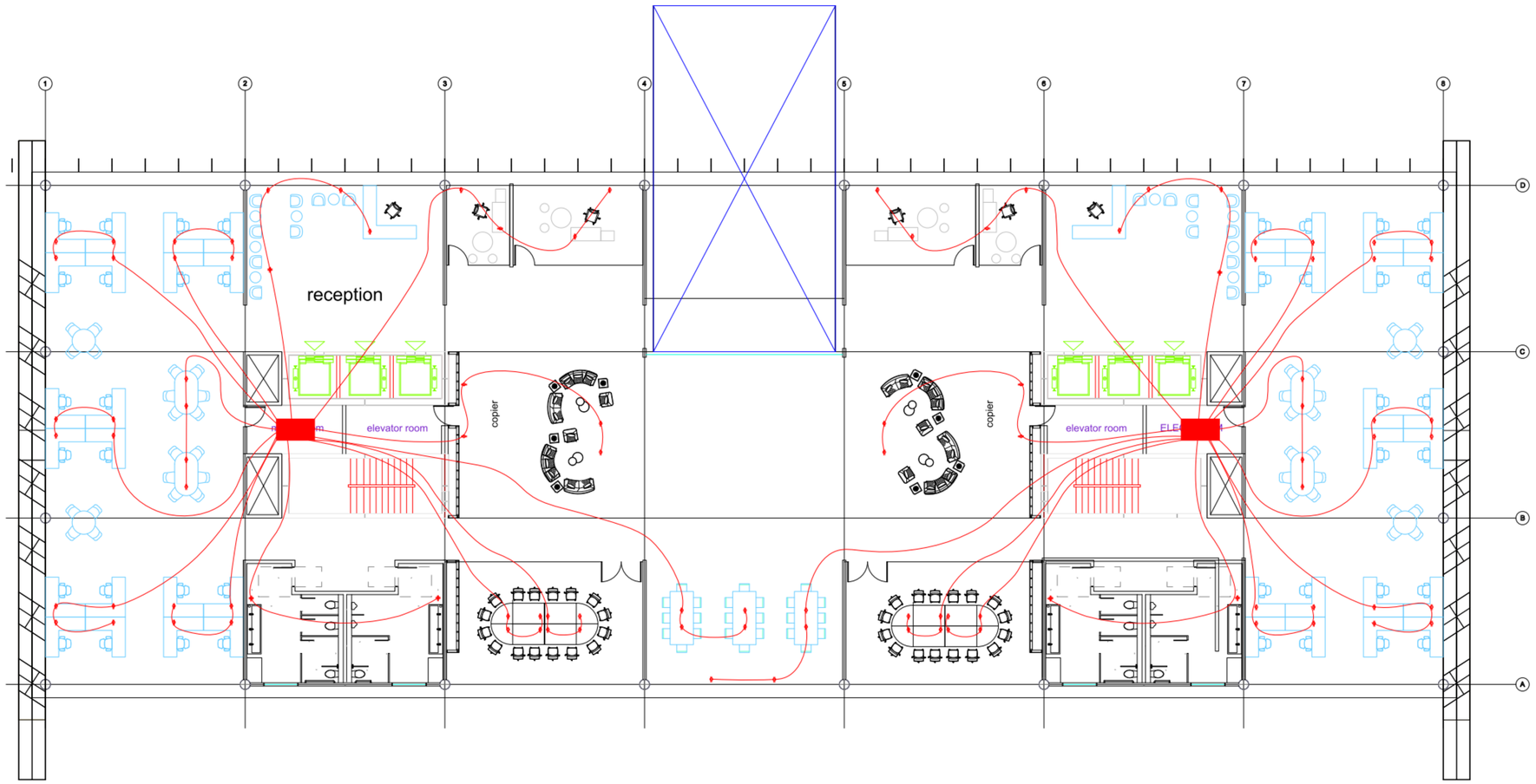
90% CD SET

PERFORMANCE ANALYSIS

Date	03.05.24
Drawn By	Author
Checked By	Checker

A601

Scale



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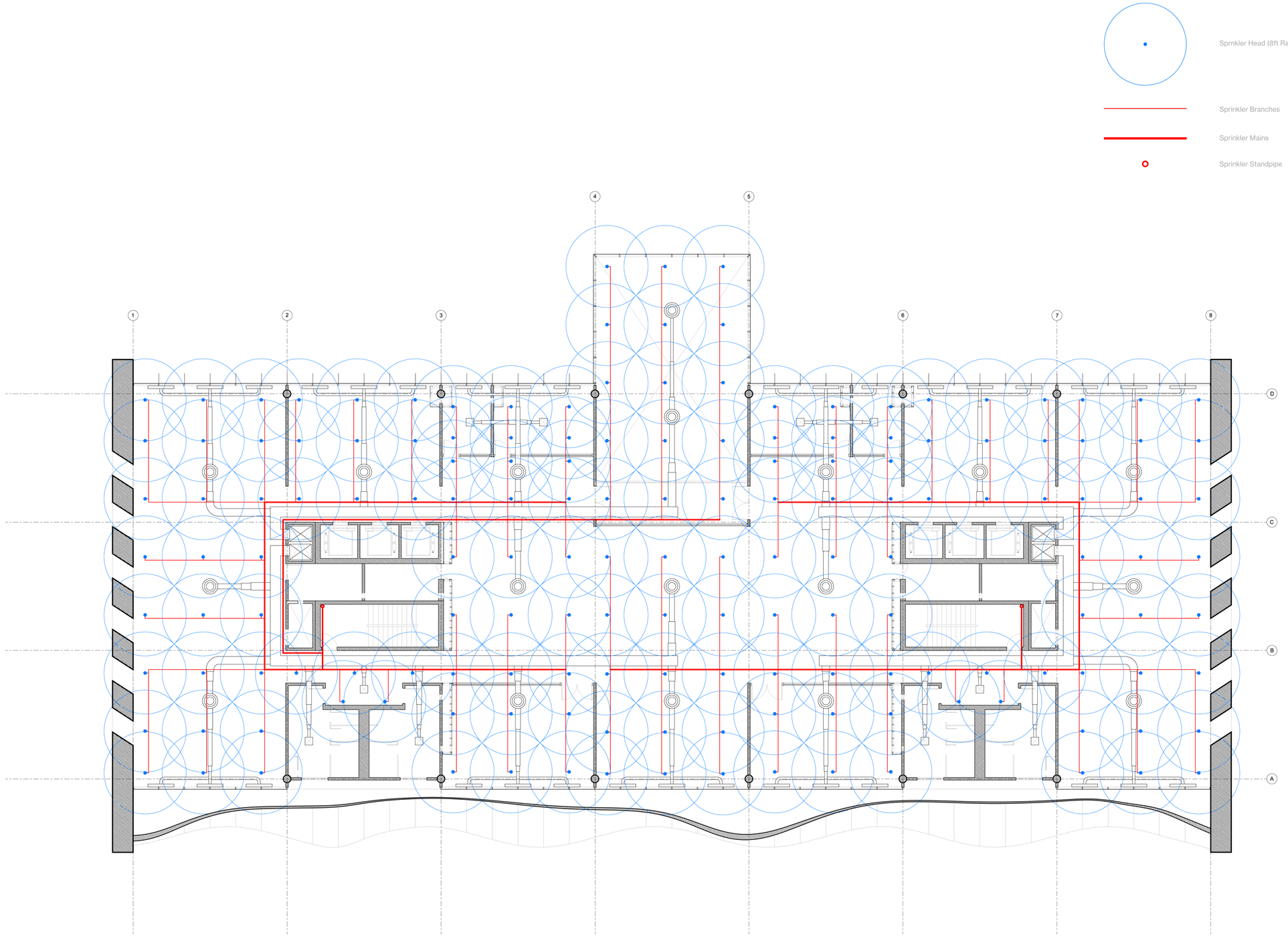
90% CD SET

ELECTRICAL POWER PLAN

Date	03.05.24
Drawn By	Author
Checked By	Checker

E100

Scale



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No.	Description	Date

90% CD SET

FIRE SAFETY PLAN

Date	03.05.24
Drawn By	Author
Checked By	Checker

FS100

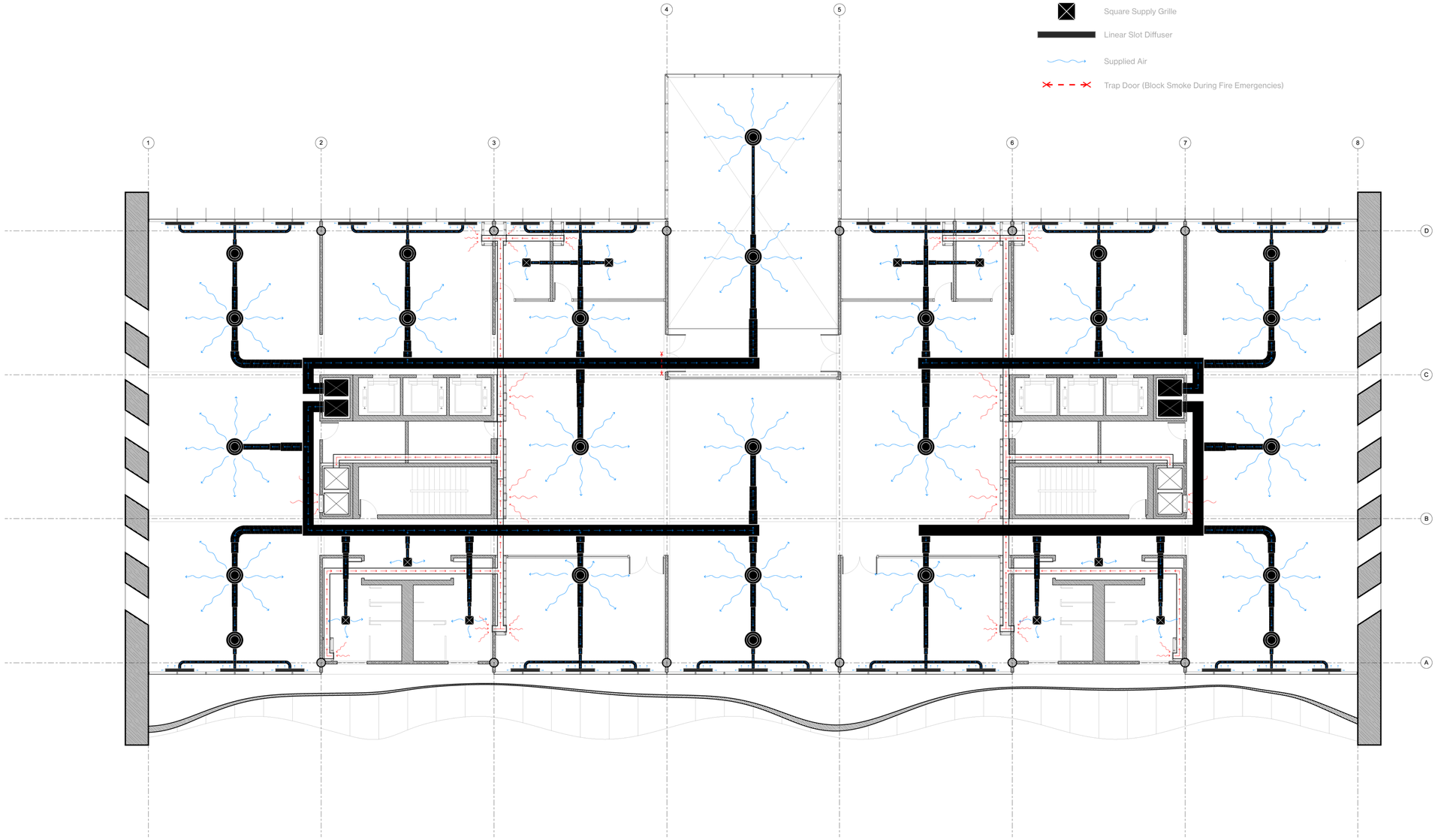
Scale

Supply

- Vertical Supply Chase
- Supply Duct (Rect.)
- Supply Duct (Circ.)
- Round Supply Grille
- Square Supply Grille
- Linear Slot Diffuser
- Supplied Air
- Trap Door (Block Smoke During Fire Emergencies)

Return

- Vertical Return Chase
- Return Duct (Rect.)
- Return Air Grilles (Mounted in wall near floor)
- Returned Air



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No.	Description	Date

90% CD SET

TYPICAL MECHANICAL PLAN

Date	03.05.24
Drawn By	Author
Checked By	Checker

M100

Scale